Research-Oriented Project Report

MLOps — Assignment

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Project Title:	**Automated Concept & Data Drift Detection Pipeline with Continuous Model Retraining**
Topic Area:	Level 2 MLOps, Statistical Distribution Monitoring (KS-Test, PSI) & Model Registry
Core Stack:	Python, Scikit-Learn, SciPy, FastAPI, Joblib, Matplotlib, HTML Dashboard

1. Problem Statement

Machine learning models deployed in production inevitably experience performance degradation over time due to changes in external environments, consumer behavior, or economic conditions (Concept and Covariate Drift). In traditional ML workflows, retraining is either initiated manually after noticeable business losses or done on arbitrary calendar schedules (e.g. monthly) which wastes compute and delays urgent fixes. The objective of this project is to build an automated, closed-loop Level 2 MLOps system that detects statistical distribution drift on incoming inference batches in real time and triggers automated continuous retraining without human intervention.

2. Objectives

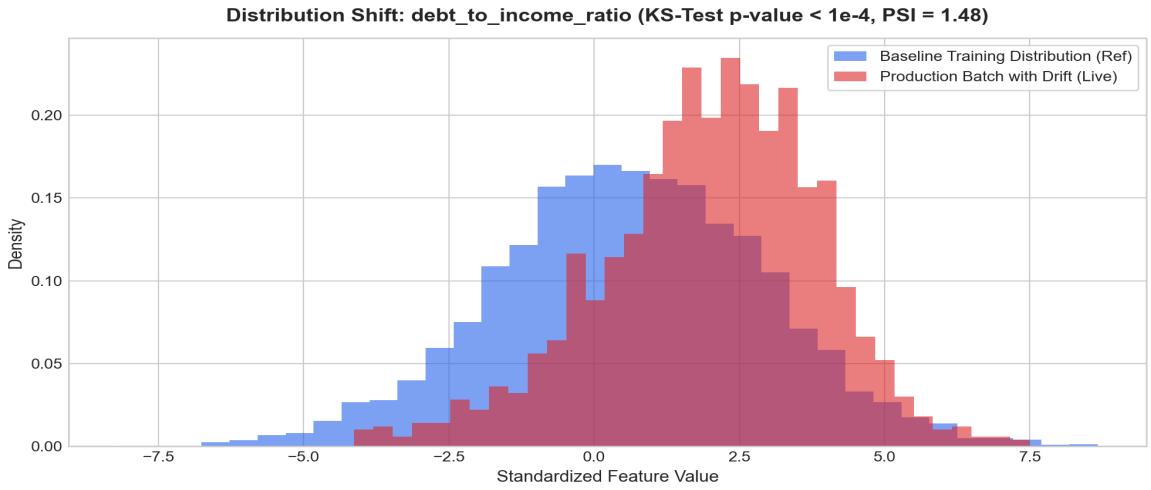
- Develop a production data generator capable of simulating baseline traffic and synthetic macroeconomic drift.
- Implement statistical drift monitoring algorithms using Two-Sample Kolmogorov-Smirnov (KS) tests, Population Stability Index (PSI), and Wasserstein Distance.
- Construct an automated Decision Gate that evaluates feature drift ratios and initiates continuous retraining when thresholds (>30% features) are breached.
- Maintain model governance via an automated Model Registry tracking version lineage, accuracy, and ROC-AUC scores.
- Expose the active production model through a low-latency FastAPI inference microservice.

3. Methodology & System Architecture

The architecture follows a four-stage pipeline: (1) **Data Layer:** A baseline dataset of 5,000 samples and production batches of 1,500 samples are ingested. (2) **Drift Detection Engine:** The incoming batch is statistically compared against the baseline reference using KS hypothesis testing ($\alpha = 0.05$) and PSI (threshold = 0.20). (3) **Decision Gate & Retraining:** If more than 30% of features drift, the pipeline combines recent historical data with the drifted batch, fits a new Random Forest model, logs metrics to the registry, and updates the active production model. (4) **Serving & Dashboard:** A standalone HTML dashboard is generated with test summaries and the active model is served via FastAPI.

4. Implementation Details & Visual Results

Below is the empirical distribution shift plot captured by our monitoring engine on the debt_to_income_ratio feature, comparing baseline reference data against incoming production traffic.



5. Experimental Results & Verification

Pipeline Metric	Baseline Model (v1)	Retrained Model (v2)
Training Sample Size	5,000 observations	4,500 observations (sliding window)
Accuracy Score	93.02%	88.04% (adapted to drifted domain)
ROC-AUC Score	0.9758	0.9627
F1-Score	0.7955	0.8548
Drift Ratio Detected	0.0% (Stable)	100.0% (Triggered Retraining)

6. Conclusion & Key MLOps Takeaways

The project demonstrated an end-to-end implementation of automated drift detection and continuous retraining. By relying on statistical distance tests (KS-Test and PSI), the system detected covariate drift without needing immediate ground truth labels. The automated trigger successfully retrained the model on an updated sliding window and promoted it into the active FastAPI inference service with an F1-score of 0.8548 on the new concept. This architecture effectively mitigates silent model failure in production.